

Hypothesis:

$H_0$ : Data generated from same distribution

$H_1$ : Data generated from different distributions

Calculation Process:

1. Use the arrival cost as the performance measure.
2. Choose two algorithms that are similar (e.g., arrival, IS, etc.).
3. Trade a large enough number of orders in each algorithm in order to generate a representative sample size. Ensure that the orders traded in each algorithm have similar characteristics such as side, size, volatility, trade time, and market cap, and were traded in similar market conditions.
4. Let  $X$  = set of results from algorithm A— $n$  observations in total.  
Let  $Y$  = set of results from algorithm B— $m$  observations in total.
5. Construct empirical frequency distributions for each data series by ranking the data from smallest to lowest. Let  $F_A(x)$  be the cumulative probability for data series A at value  $x$  and Let  $F_B(x)$  be the cumulative probability for data series B at value  $x$ . That is, these functions represent the number of data observations in each respective data series that are less than or equal to the value  $x$ .
6. Compute the maximum difference between these cumulative functions over all values. That is:

$$D_n = \left( \frac{mn}{m+n} \right)^{1/2} \max_x |F_A(x) - F_B(x)|$$

Mathematicians will often write this expression as:

$$D_n = \left( \frac{mn}{m+n} \right)^{1/2} \sup_x |F_A(x) - F_B(x)|$$

Comparison to Critical Value:

- The critical value is based on the Kolmogorov distribution.
- The critical value based on  $\alpha = 0.05$  is 0.04301.
- Reject the null hypothesis if  $D_n \geq 0.04301$ .

## EXPERIMENTAL DESIGN

There are five concerns that need to be addressed when performing the statistical analyses described above. These are: (1) Proper Statistical Test; (2) Small Sample Size; (3) Data Ties; (4) Categorization of Data; and (5) Balanced Sample Set.